



PES UNIVERSITY

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Department of Computer Science & Engineering

Session: AUG-DEC 2025

UE24MA242A: Mathematics for Computer Science Engineers

Classwork Problems

#	UNIT_3 Questions	Exercise Number
1	In an experiment to measure the durability of cooling fans used in high-performance computer servers, 73 fans were tested until failure under continuous operation. The mean lifetime of the sample was 783 hours, and the standard deviation was 120 hours. Let μ represent the mean lifetime (in hours) of all such cooling fans produced by the company. A test is made of $H_0: \mu \leq 750$ versus $H_1: \mu > 750$. a) Find the P-value. b) Either the mean number of kilocycles to failure is greater than 750, or the sample is in the most extreme _____ % of its distribution.	Testing for population mean
2	A certain type of stainless steel powder is supposed to have a mean particle diameter of $\mu = 15\mu\text{m}$. A random sample of 87 particles had a mean diameter of $15.2\mu\text{m}$, with a standard deviation of $1.8\mu\text{m}$. A test is made of $H_0: \mu = 15$ versus $H_1: \mu$ not equal to 15. a) Find the P-value. b) Do you believe it is plausible that the mean diameter is $15\mu\text{m}$, or are you convinced that it differs from $15\mu\text{m}$? Explain your reasoning.	Section 6.1 Q. No: 6
3	For which P-value is the null hypothesis more plausible: $P = 0.5$ or $P = 0.05$?	Section 6.2 Q. No: 1
4	True or false: a) If we reject H_0, then we conclude that H_0 is false. b) If we do not reject H_0, then we conclude that H_0 is true. c) If we reject H_0, then we conclude that H_1 is true. d) If we do not reject H_0, then we conclude that H_1 is false.	Section 6.2 Q. No: 2

5	If $P = 0.01$, which is the best conclusion? a) H_0 is definitely false. b) H_0 is definitely true. c) There is a 1% probability that H_0 is true. d) H_0 might be true, but it's unlikely. e) H_0 might be false, but it's unlikely. f) H_0 is plausible.	Section 6.2 Q. No: 3
6	If $P = 0.50$, which is the best conclusion? a) H_0 is definitely false. b) H_0 is definitely true. c) There is a 50% probability that H_0 is true. d) H_0 is plausible, and H_1 is false. e) Both H_0 and H_1 are plausible.	Section 6.2 Q. No: 4
7	Let μ be the radiation level to which a radiation worker is exposed during the course of a year. The Environmental Protection Agency has set the maximum safe level of exposure at 5 rems per year. If a hypothesis test is to be performed to determine whether a work place is safe, which is the most appropriate null hypothesis: $H_0: \mu \leq 5$, $H_0: \mu \geq 5$, or $H_0: \mu = 5$? Explain.	Section 6.2 Q. No: 8
8	A random sample of 300 electronic components manufactured by a certain process are tested, and 25 are found to be defective. Let p represent the proportion of components manufactured by this process that are defective. The process engineer claims that $p \leq 0.05$. Does the sample provide enough evidence to reject the claim?	Section 6.3 Q. No: 1
9	In a survey of 500 residents in a certain town, 274 said they were opposed to constructing a new shopping mall. Can you conclude that more than half of the residents in this town are opposed to constructing a new shopping mall?	Section 6.3 Q. No: 5
10	A new centrifugal pump is being considered for an application involving the pumping of ammonia. The specification is that the flow rate be more than 5 gal lons per minute (gpm). In an initial study, eight runs were made. The average flow rate was 6.5gpm and the standard deviation was 1.9gpm. If the mean flow rate is found to meet the specification, the pump will be put into service. a. State the appropriate null and alternate hypotheses. b. Find the P-value. c. Should the pump be put into service? Explain.	Section 6.4 Q. No: 3
11.	The thicknesses of eight pads designed for use in air craft engine mounts are measured. The results, in mm, are 41.83, 41.01, 42.68, 41.37, 41.83, 40.50, 41.70, and 41.42. Assume that the thicknesses are a sample from an approximately symmetric distribution. a. Can you conclude that the mean thickness is greater than 41 mm? Compute the appropriate test statistic and find the P-value. b. Can you conclude that the mean thickness is less than 41.8 mm? Compute the appropriate test statistic and find the P-value. c. The target thickness is 42 mm. Can you conclude that the mean thickness differs from the target value? Compute the appropriate test statistic and find the P-value.	Section 6.9 Q. No: 2

12.	The article “n-Nonane Hydroconversion on Ni and Pt Containing HMFI, HMOR and HBEA” (G. Kinger and H. Vinek, Applied Catalysis A:General,2002:139–149) presents hydro conversion rates(in $\mu\text{mol/g}\cdot\text{s}$)of n-nonane over both HMFI and HBEA catalysts. The results are as follows: HMFI: 0.43 0.93 1.91 2.56 3.72 6.19 11.00 HBEA: 0.73 1.12 1.24 2.93 Can you conclude that the mean rate differs between the two catalysts?	Section 6.9 Q. No: 6																				
13.	In a comparison of the effectiveness of distance learning with traditional classroom instruction,12 students took a business administration course online, while 14 students took it in a classroom. The final exam scores were as follows. Online: 66 75 85 64 88 77 74 91 72 69 77 83 Classroom: 80 83 64 81 75 80 86 81 51 64 59 85 74 77 Can you conclude that the mean score differs between the two types of course?	Section 6.9 Q. No: 9																				
14.	At an assembly plant for light trucks, routine monitoring of the quality of welds yields the following data: <table><tr><th></th><th colspan="3">Number of Welds</th></tr><tr><th></th><th>High Quality</th><th>Moderate Quality</th><th>Low Quality</th></tr><tr><td>Day Shift</td><td>467</td><td>191</td><td>42</td></tr><tr><td>Evening Shift</td><td>445</td><td>171</td><td>34</td></tr><tr><td>Night Shift</td><td>254</td><td>129</td><td>17</td></tr></table> Can you conclude that the quality varies among shifts? a. State the appropriate null hypothesis. b. Compute the expected values under the null hypothesis. c. Compute the value of the chi-square statistic. d. Find the P-value. What do you conclude?		Number of Welds				High Quality	Moderate Quality	Low Quality	Day Shift	467	191	42	Evening Shift	445	171	34	Night Shift	254	129	17	Section 6.10 Q. No: 2
	Number of Welds																					
	High Quality	Moderate Quality	Low Quality																			
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15	For the given table of observed values, a. Construct the corresponding table of expected values. b. If appropriate, perform the chi-square test for the null hypothesis that the row and column outcomes are independent. If not appropriate, explain why. <table><tr><th></th><th colspan="3">Observed Values</th></tr><tr><th></th><th>1</th><th>2</th><th>3</th></tr><tr><td>A</td><td>15</td><td>10</td><td>12</td></tr><tr><td>B</td><td>3</td><td>11</td><td>11</td></tr><tr><td>C</td><td>9</td><td>14</td><td>12</td></tr></table>		Observed Values				1	2	3	A	15	10	12	B	3	11	11	C	9	14	12	Section 6.10 Q. No: 7
	Observed Values																					
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C	9	14	12																			

16.	A process for a certain type of ore is designed to reduce the concentration of impurities to less than 2%. It is known that the standard deviation of impurities for processed ore is 0.6%. Let μ represent the mean impurity level, in percent, for ore specimens treated by this process. The impurity of 80 ore specimens is measured, and a test of the hypothesis $H_0: \mu \geq 2$ versus $H_1: \mu < 2$ will be performed. a. If the test is made at the 5% level, what is the rejection region? b. If the sample mean impurity level is 1.85, will H_0 be rejected at the 10% level? c. If the sample mean pH is 1.85, will H_0 be rejected at the 1% level? d. If the value 1.9 is a critical point, what is the level of the test?	Section 6.12 Q. No: 2																																																
17.	A test is made of the hypotheses $H_0: \mu \leq 10$ versus $H_1: \mu > 10$. For each of the following situations, determine whether the decision was correct, a type I error occurred, or a type II error occurred. a. $\mu = 8$, H_0 is rejected. b. $\mu = 10$, H_0 is not rejected. c. $\mu = 14$, H_0 is not rejected. d. $\mu = 12$, H_0 is rejected.	Section 6.12 Q. No: 4																																																
18.	A hypothesis test is to be performed, and it is decided to reject the null hypothesis if $P \leq 0.10$. If H_0 is in fact true, what is the maximum probability that it will be rejected?	Section 6.12 Q. No: 6																																																
19.	A test has power 0.90 when $\mu = 15$. True or false: a. The probability of rejecting H_0 when $\mu = 15$ is 0.90. b. The probability of making a correct decision when $\mu = 15$ is 0.90. c. The probability of making a correct decision when $\mu = 15$ is 0.10. d. The probability that H_0 is true when $\mu = 15$ is 0.10.	Section 6.13 Q. No: 1																																																
20.	Compute the correlation coefficient for the following data set. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>2</td><td>1</td><td>4</td><td>3</td><td>7</td><td>5</td><td>6</td></tr></table>	x	1	2	3	4	5	6	7	y	2	1	4	3	7	5	6	Section 7.1 Q. No: 1																																
x	1	2	3	4	5	6	7																																											
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21.	For each of the following data sets, explain why the correlation coefficient is the same as for the data set in the previous question. a. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>5</td><td>4</td><td>7</td><td>6</td><td>10</td><td>8</td><td>9</td></tr></table> b. <table><tr><td>x</td><td>11</td><td>21</td><td>31</td><td>41</td><td>51</td><td>61</td><td>71</td></tr><tr><td>y</td><td>5</td><td>4</td><td>7</td><td>6</td><td>10</td><td>8</td><td>9</td></tr></table> c. <table><tr><td>x</td><td>53</td><td>43</td><td>73</td><td>63</td><td>103</td><td>83</td><td>93</td></tr><tr><td>y</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td><td>16</td></tr></table>	x	1	2	3	4	5	6	7	y	5	4	7	6	10	8	9	x	11	21	31	41	51	61	71	y	5	4	7	6	10	8	9	x	53	43	73	63	103	83	93	y	4	6	8	10	12	14	16	Section 7.1 Q. No: 2
x	1	2	3	4	5	6	7																																											
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22.	True or false, and explain briefly: a. If the correlation coefficient is positive, then above-average values of one variable are associated with above-average values of the other. b. If the correlation coefficient is negative, then below-average values of one variable are associated with below-average values of the other. c. If y is usually less than x, then the correlation between y and x will be negative.	Section 7.1 Q. No: 4												
23.	In a study of ground motion caused by earthquakes, the peak velocity (inm/s) and peak acceleration (in m/s²) were recorded for five earthquakes. The results are presented in the following table. <table><tr><td>Velocity</td><td>1.54</td><td>1.60</td><td>0.95</td><td>1.30</td><td>2.92</td></tr><tr><td>Acceleration</td><td>7.64</td><td>8.04</td><td>8.04</td><td>6.37</td><td>5.00</td></tr></table> a. Compute the correlation coefficient between peak velocity and peak acceleration. b. Construct a scatter plot for these data. c. Is the correlation coefficient an appropriate summary for these data? Explain why or why not. d. Someone suggests converting the units from meters to centimetres and from seconds to minutes. What effect would this have on the correlation?	Velocity	1.54	1.60	0.95	1.30	2.92	Acceleration	7.64	8.04	8.04	6.37	5.00	Section 7.1 Q. No: 6
Velocity	1.54	1.60	0.95	1.30	2.92									
Acceleration	7.64	8.04	8.04	6.37	5.00									
24.	Each month for several months, the average temperature in °C(x) and the number of pounds of steam (y) consumed by a certain chemical plant were measured. The least-squares line computed from the resulting data is y = 245.82 +1.13x. a. Predict the number of pounds of steam consumed in a month where the average temperature is 65°C. b. If two months differ in their average temperatures by 5°C, by how much do you predict the number of pounds of steam consumed to differ?	Section 7.2 Q. No: 1												
25.	A least-squares line is fit to a set of points. If the total sum of squares is $\sum (y_i - \bar{y})^2 = 181.2$, and the error sum of squares is $\sum (y_i - \hat{y}_i)^2 = 33.9$, compute the coefficient of determination r^2.	Section 7.2 Q. No: 4												
26.	The processing of raw coal involves “washing,” in which coal ash (nonorganic, incombustible material) is removed. The article “Quantifying Sampling Precision for Coal Ash Using Gy’s Discrete Model of the Fundamental Error (Journal of Coal Quality, 1989:33–39) provides data relating the percentage of ash to the density of a coal particle. The average percentage ash for five densities of coal particles was measured. The data are presented in the following table: <table><tr><td>Density (g/cm³)</td><td>Percent ash</td></tr><tr><td>1.25</td><td>1.93</td></tr><tr><td>1.325</td><td>4.63</td></tr><tr><td>1.375</td><td>8.95</td></tr><tr><td>1.45</td><td>15.05</td></tr><tr><td>1.55</td><td>23.31</td></tr></table>	Density (g/cm ³)	Percent ash	1.25	1.93	1.325	4.63	1.375	8.95	1.45	15.05	1.55	23.31	Section 7.2 Q. No: 10
Density (g/cm ³)	Percent ash													
1.25	1.93													
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1.55	23.31													

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| | <ul style="list-style-type: none">a. Construct a scatterplot of percent ash (y) versus density (x). Verify that a linear model is appropriate.b. Compute the least-squares line for predicting percent ash from density.c. If two coal particles differed in density by 0.1g/cm^3, by how much would you predict their percent ash to differ?d. Predict the percent ash for particles with density 1.40g/cm^3.e. Compute the fitted values.f. Compute the residuals. Which point has the residual with the largest magnitude?g. Compute the correlation between density and percent ash.h. Compute the regression sum of squares, the error sum of squares, and the total sum of squares.i. Divide the regression sum of squares by the total sum of squares. What is the relationship between this quantity and the correlation coefficient? | |
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